Famicom JAMMA board

power

File: power.kicad_sch

Power: +5V entry, switched +5V_SW, +3V3, supply of the RP2A03 modules

cpu

File: cpu.kicad_sch

CPU: RP2A03, work RAM, address decoder, data-bus resistors

ppu

File: ppu.kicad_sch

PPU: RP2C02, address latch, video RAM, sprite detection

cart

File: cart.kicad_sch

Cartridge connector and the dump circuit for emulation mode

video

File: video.kicad_sch

Video: RGB resistor DAC and composite sync

audio

File: audio.kicad_sch

Audio: mixer, loop through the cartridge, power amplifier

jamma

File: jamma.kicad_sch

JAMMA edge connector, inputs, coin counter and lockout drivers

mcu

File: mcu.kicad_sch

Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

PWR_FLAG

GND

PWR_FLAG

+5V

PWR_FLAG

+5V_SW

PWR_FLAG

VSYS

Sheet: /
File: fcjamma.kicad_sch

Title: Famicom JAMMA board

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 1/9

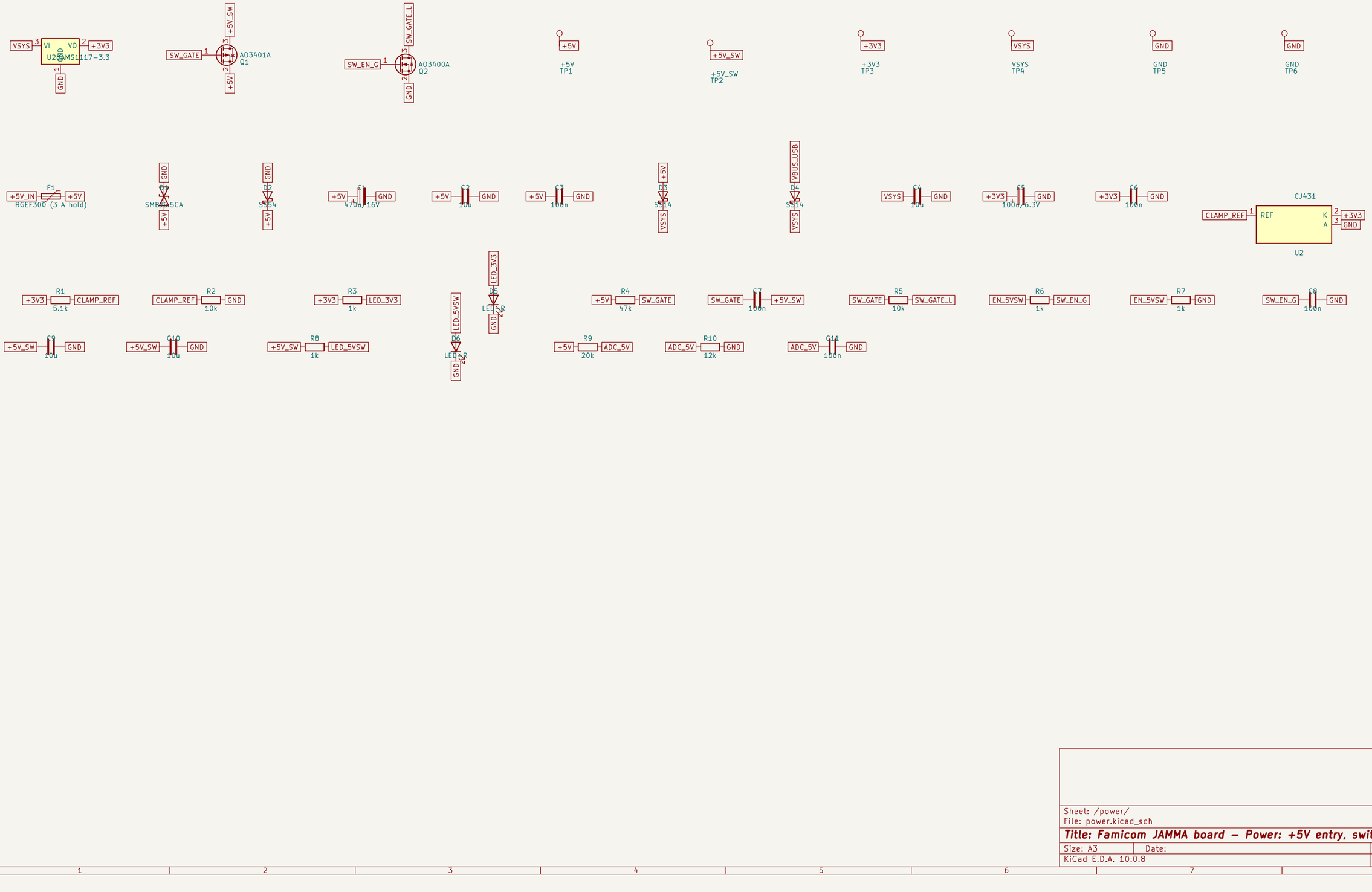
Power: +5V entry, switched +5V_SW, +3V3, supply of the Pico modules

+5V_IN comes from the JAMMA fingers 3, 4, C, D. −5 V and +12 V are not used.

+5V_SW feeds the CPU, the PPU, the memories, the 5 V logic and the cartridge. It is off until Pico C raises EN_5VSW,

so that no 5 V signal reaches a Pico pin before that Pico is powered (the RP2350 pins tolerate 5 V only while powered).

VSYS feeds the three Pico modules and the +3V3 regulator from +5V or from the USB socket, whichever is present.



Sheet: /power/
File: power.kicad_sch

Title: Famicom JAMMA board — Power: +5V entry, switched +5V_SW, +3V3, s

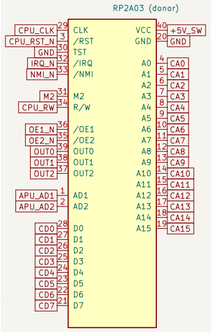
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 2/9

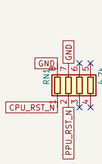
CPU: RP2A03, work RAM, address decoder, data-bus resistors

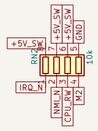
U1 is the donor's RP2A03 in a 40-pin socket. In emulation mode the socket is empty and the dump shift registers drive the buses.

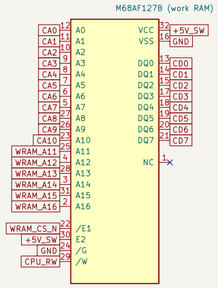
The 470 ohm resistors let Pico B replace what the PPU sees on a write without disturbing the CPU side of the bus.

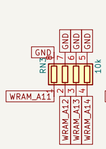
No pull-ups on the CPU data bus: they are known to upset some PPU revisions, and games read the floating bus.

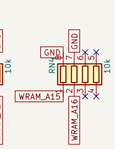


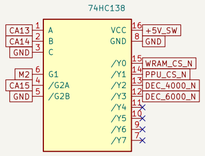


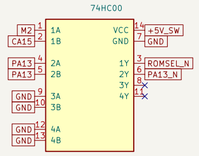


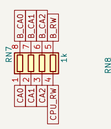


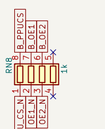


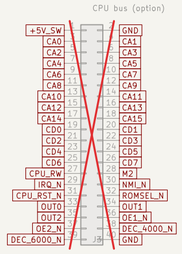


































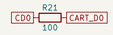


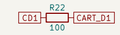




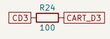
























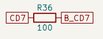




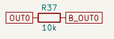




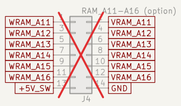












PPU: RP2C02, address latch, video RAM, sprite detector

U6 is the donor's RP2C02 in a 40-pin socket. EXT0–3 give the lower four bits of the palette index of every pixel once Pico B has forced bit 6 of register 2000 high. The fifth bit (sprite or background) is read from the brightness of VOUT: Pico B makes the PPU store a dark colour in every background entry and a bright one in every sprite entry.

RP2C02 (donor)

U6

74HC373

U7

M68AF127B (video RAM)

U8

TL712CP

U9

PPU bus (option)

J4

VBUF
TP9

VREF
TP10

CMP_OUT
TP11

MMBT3904
Q3

composite video (test)

J5

sprite bit source

J6

PPU /RST = +5V (Famicom PPU)

JP1

Sheet: /ppu/	
File: ppu.kicad_sch	
Title: Famicom JAMMA board – PPU: RP2C02, address latch, video RAM, sprit	
Size: A2	Date:
KiCad E.D.A. 10.0.8	Rev: 0.1
	Id: 4/9

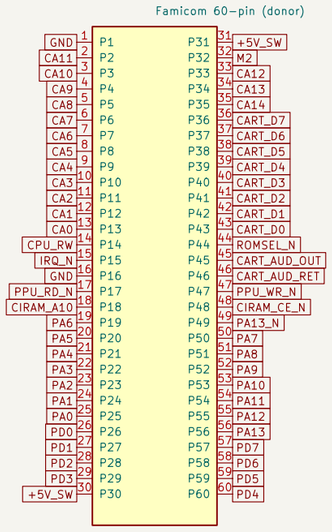
Cartridge connector and the dump circuit for emulation mode

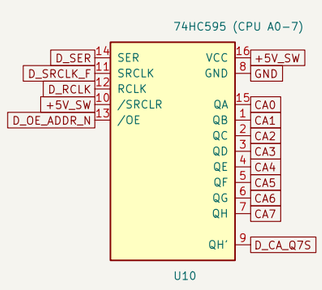
J8 is the 60-pin connector taken from the donor Famicom. Pin 45 carries the board's sound to the cartridge, pin 46 brings it back.

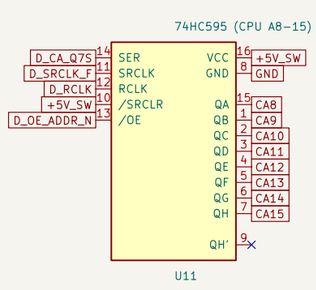
Dump circuit: with the CPU and PPU sockets empty, Pico C reads the cartridge through shift registers and hands the data to Pico A.

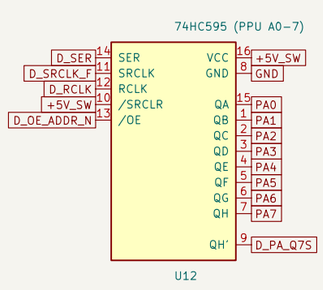
The six 74HC595 run from +5V_SW; Pico C drives their inputs open-drain against 4.7 k pull-ups, so they see proper 5 V levels.

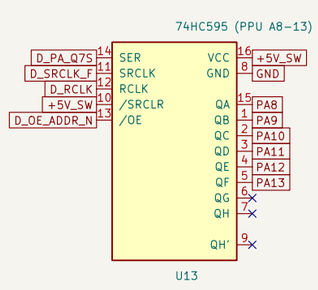
In real-chip mode D_OE_ADDR_N and D_OE_DATA_N stay high (pull-ups) and every 74HC595 output is switched off.

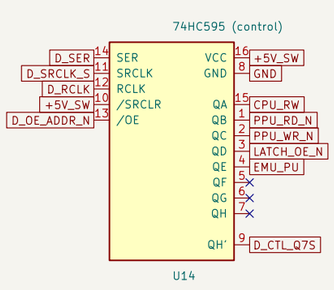


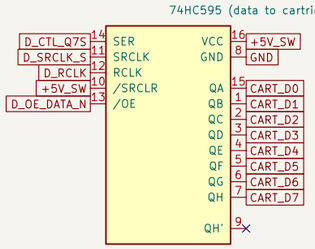


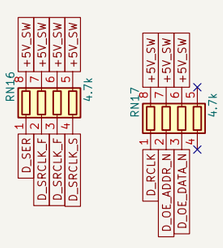


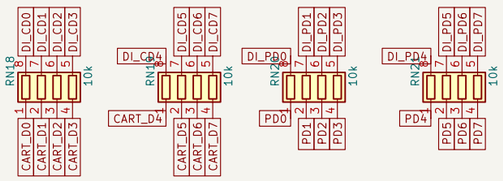


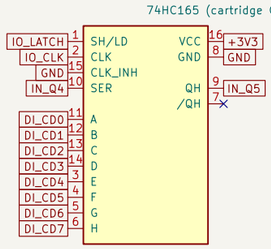


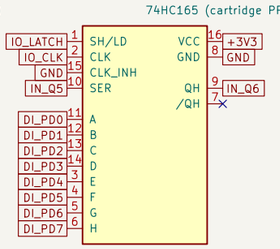
















Sheet: /cart/
File: cart.kicad_sch

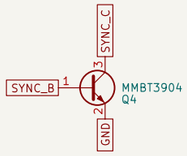
Title: Famicom JAMMA board – Cartridge connector and the dump circuit for

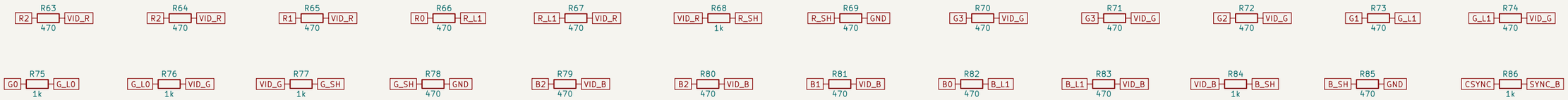
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 5/9

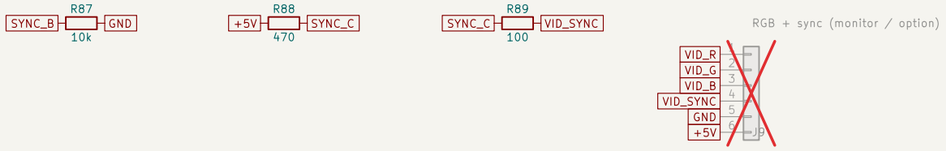
Video: RGB resistor DAC and composite sync

Pico A drives the DAC directly. Full scale is 3.0 V open circuit, about 2.85 V into 2 k, source resistance about 125 ohm.

Red and blue have 3 bits, green has 4. Only 470 ohm and 1 k are used so that the parts list stays short.







Sheet: /video/
File: video.kicad_sch

Title: Famicom JAMMA board – Video: RGB resistor DAC and composite sync

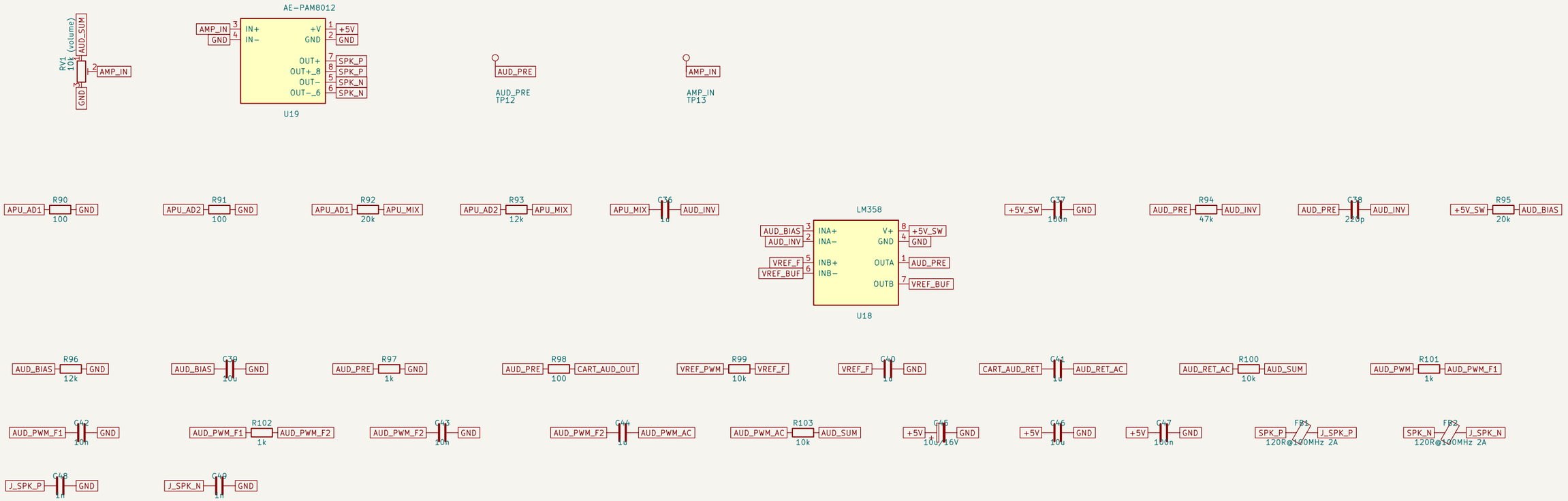
Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 6/9

Audio: mixer, loop through the cartridge, power amplifier

Real-chip mode: 2A03 pins -> mixer (Famicom values) -> cartridge pin 45 -> cartridge -> pin 46 -> volume -> amplifier.

Emulation mode: Pico A's PWM output -> two RC stages -> volume -> amplifier.

The loudspeaker outputs are bridge-tied: JAMMA L is not ground.



Sheet: /audio/		
File: audio.kicad_sch		
Title: Famicom JAMMA board – Audio: mixer, loop through the cartridge, pow		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 7/9

JAMMA edge connector, inputs, coin counter and lock-out drivers

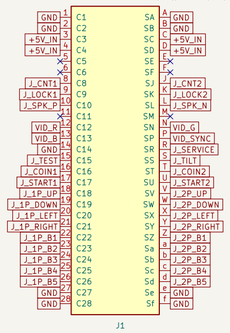
Every input: 1 k pull-up to +3V3 (3.3 mA through the switch), 4.7 k in series, 10 nF to ground, then a 74HC165 on +3V3.

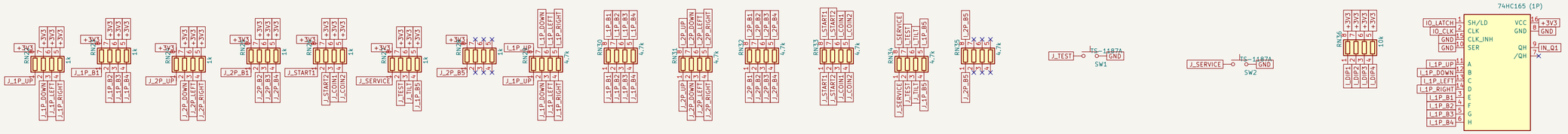
Pico C reads the chain: after a low pulse on IO_LATCH the first bit on IN_DATA is input H of U17, the last is input A of U20.

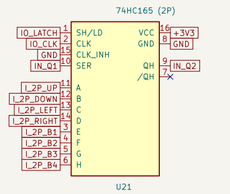
The same pulse copies the 74HC595's shift register to its outputs, so one cycle reads the inputs and writes the outputs.

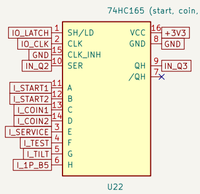
Fingers 5/E (−5 V), 6/F (+12 V) and 11/M are not connected. 7/H is the key slot.

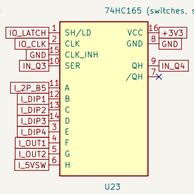
JAMMA edge (gold fingers)

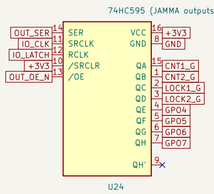


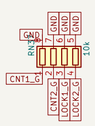


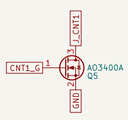


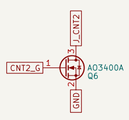


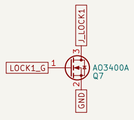


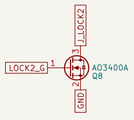


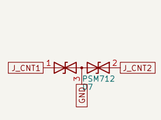


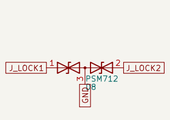


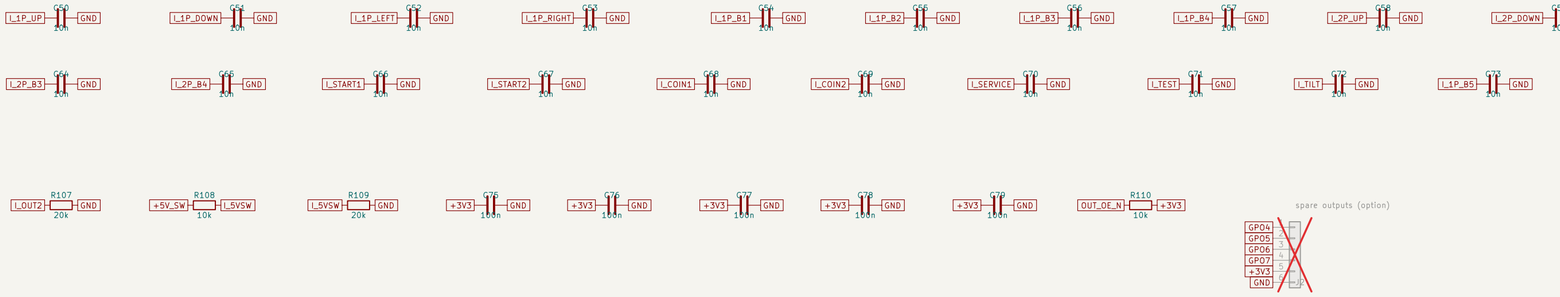


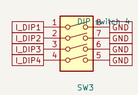


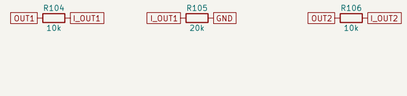












Sheet: /jamma/
File: jamma.kicad_sch

Title: Famicom JAMMA board — JAMMA edge connector, inputs, coin counter a

Size: A2	Date:	**Rev: 0.1**
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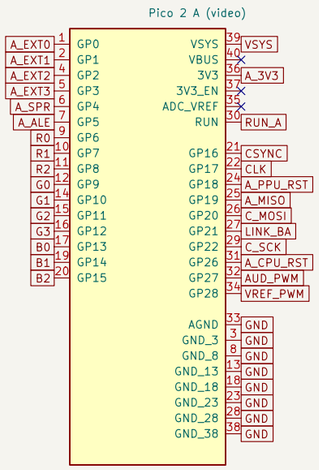
Three Raspberry Pi Pico 2 modules, USB Type-C socket, links

Pico A (video): clock and reset for CPU and PPU, palette index and sprite bit in, RGB and sync out; the emulator in emulation mode.

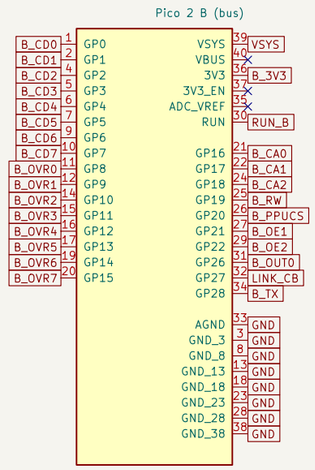
Pico B (bus): watches the CPU's accesses to the PPU, replaces palette data on the PPU side, answers the controller reads.

Pico C (input/output and management): JAMMA inputs and outputs, coin and timer logic, cartridge dump, USB Type-C,

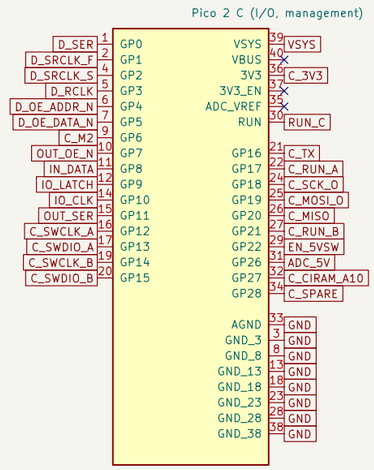
power switch of the Famicom half, and reprogramming of Pico A and Pico B through their debug pins (J13, J14).





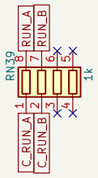


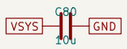


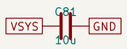


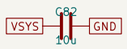








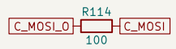


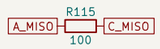




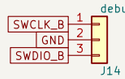


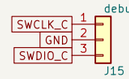




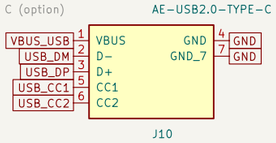


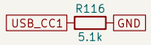


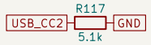


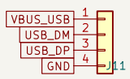












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Title: Famicom JAMMA board – Three Raspberry Pi Pico 2 modules, USB Type

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